	Pimpri Chinchwad Education Trust's **Pimpri Chinchwad College of Engineering** **(An Autonomous Institute Affiliated to SPPU,Pune)**
FA-2 – Case Study based Assignment Guidelines	

Department: Computer Engineering **Academic Year: 2025- 2026** **Semester: II**

Name of Course: DBMS **Class :** SY Computer

Formative Assessment (FA):

Marks	Activity name and its rubric	Syllabus units or topics (on which contents FA related activity is being planned)	Date or Duration of submission of activity (you may refer event calendar)
20	**Case Study based Assignment Presentation**	Unit no 4 and 5	Second Week of April 2026

Case Study based Assignment presentation on Transaction and NOSQL database:

Team Formation: Same teams of 2 students as FA-1 (continuity maintained).

Problem Statement: Continue with the mini project problem statement selected in FA-1. Extend it to transaction management and NoSQL implementation.

- Analyze transaction schedules related to the chosen mini project.
- Check conflict serializability and view serializability with proper justification.
- Ensure ACID properties (Atomicity, Consistency, Isolation, and Durability) by demonstrating understanding of concurrency control and recovery concepts (commit, rollback, and deadlock scenarios).
- Design equivalent collections for the mini project problem statement.
- Write at least 10 MongoDB queries covering CRUD operations, aggregation, indexing, and search features.
- Understanding of Comparison of relational SQL queries and MongoDB queries

Rubrics for evaluation:

Criteria	Performance Levels	Marks
1. **Schedule, Conflict & View Serializability** (CO4)	**Excellent (5):** Correct schedule analysis, accurate conflict & view serializability with clear justification **Good (4):** Minor mistakes in analysis or justification. **Basic (2–3):** Partial understanding, incomplete or unclear justification. **Poor (0–1):** Incorrect or not attempted.	**5**
2. **ACID, Concurrency Control & Recovery** (CO4)	**Excellent (5):** All ACID properties explained with examples; clear understanding of commit, rollback, deadlock & recovery. **Good (4):** Minor gaps in concepts or examples. **Basic (2–3):** Limited explanation, missing some concepts. **Poor (0–1):** Incorrect or missing.	**5**
3. **MongoDB Queries Implementation** (CO5)	**Excellent (9–10):** Well-designed collections, ≥10 correct queries (CRUD, aggregation, indexing, search), clear understanding of SQL vs MongoDB comparison. **Good (7–8):** Minor errors in queries with clear understanding **Basic (4–6):** Limited queries or partial understanding comparison. **Poor (0–3):** Incorrect, insufficient, or missing work.	**10**

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